Subject Code: CY1L001	Subject Name: Chemistry	L-T-P: 3-1-0	Credit: 4

Functional Materials - Design & Application:

(a) Synthetic Polymers (carbon framework, silicon framework, fluorinated polymer), Bio & biodegradable polymers.

(b) Surfactants.

(c) Nanostructures, Soft materials and Thin Films.

(b) Emerging applications in Energy harvesting, Memory Storage and Micro-fabrication.

Industrial & Bio-inspired Chemistry:

(a)Case studies on Industrial organics with emphasis to Drugs

(b) Oxidation, Reduction, Catalytic hydrogenation and Electron transfer.

Molecules in Daily Life: A short tour on molecules behind taste, smell, pain, colour and sex.

Reference Books:

Advanced Organic Chemistry: Reactions, Mechanisms, and Structure, by Jerry March

Morrison R. T. and Boyd R. N. *Organic Chemistry*, Prentice Hall.

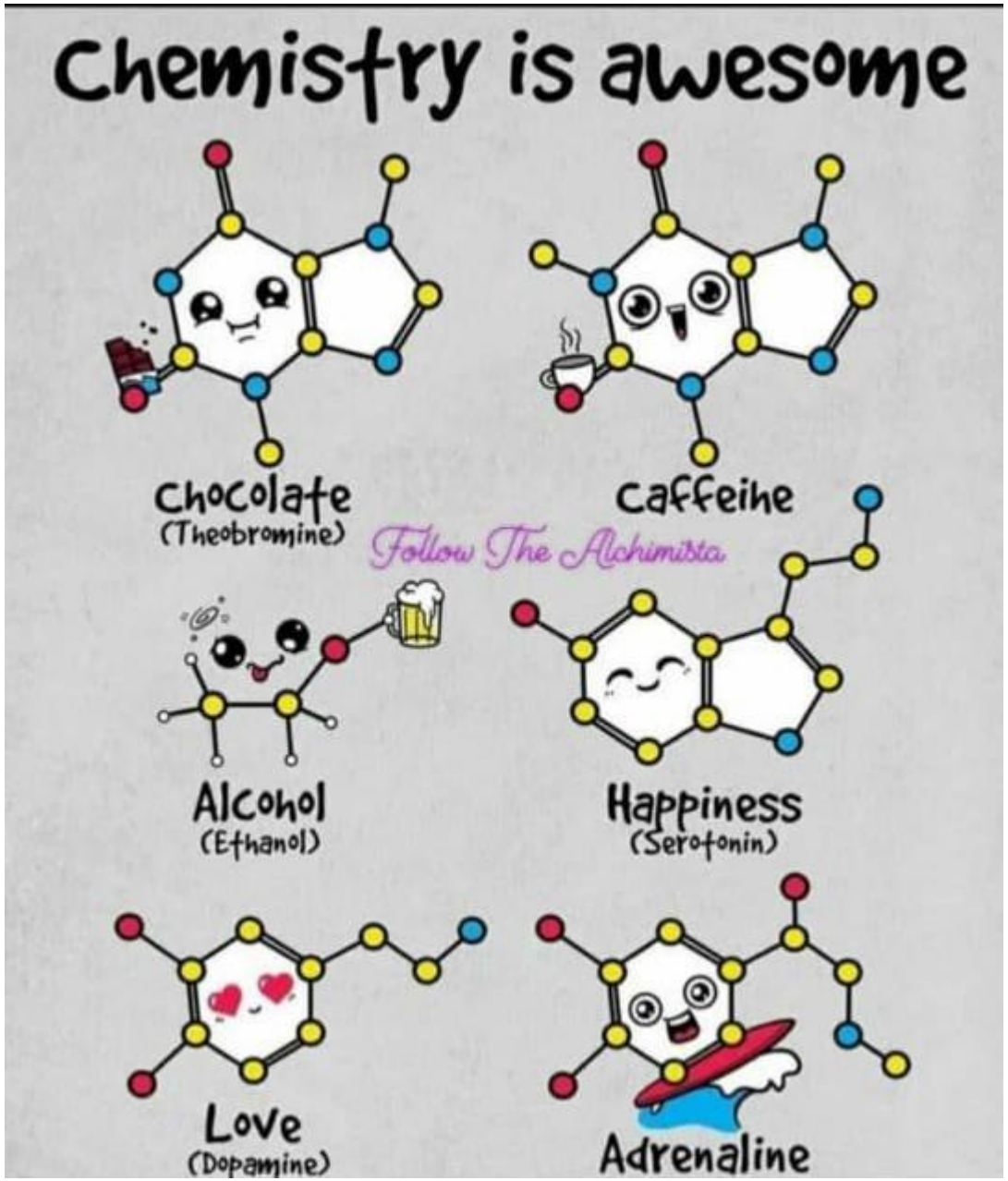
"Principles of Polymerization" by George Odian

Molecules in Daily Life: A short tour on molecules behind taste, smell, pain, colour and sex

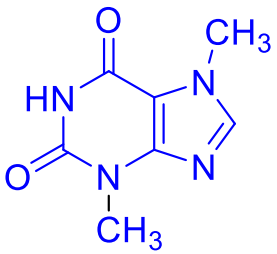
Dr. Vijayakrishna Kari
Associate Professor – Chemistry
School of Basic Sciences
IIT Bhubaneswar

Chemistry is a big part of our everyday life. We find chemistry in daily life in the foods we eat, the air we breathe, our soap, our emotions and literally every object we can see or touch. Here's a look at some everyday chemistry.

Many of the changes we observe in the world around us are caused by chemical reactions.

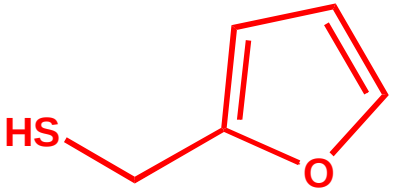


CNS (Central Nervous System) stimulant and Respiratory stimulant



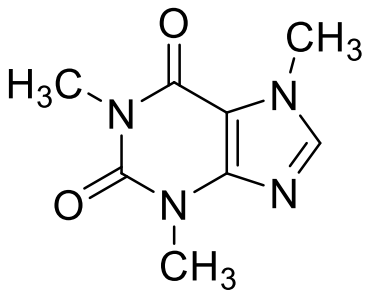
Theobromine (Xantheose)

Present in Coco



FURFURYL MERCAPTAN

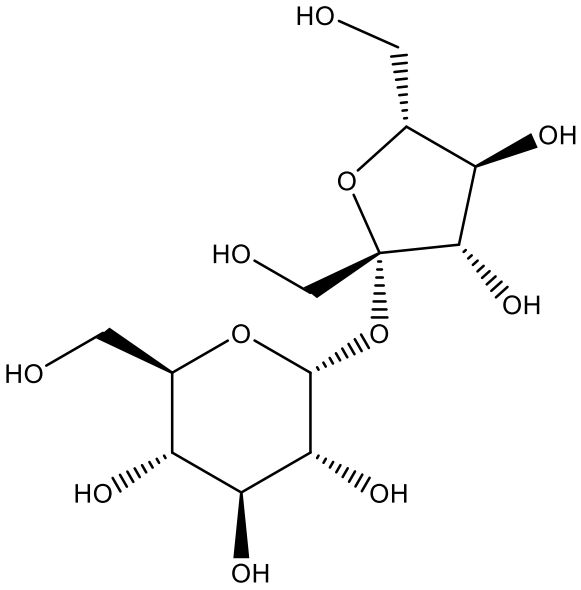
Coffee aroma



CAFFEINE

Present in Coffee

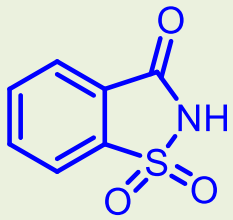
Naturally occurring alkaloid commonly found in tea leaves, coffee beans, Coco beans, originally isolated from coffee in 1821.



Sucrose

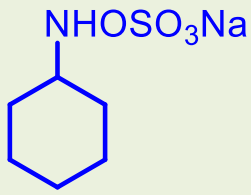
Artificial Sweeteners

Non caloric sweeteners



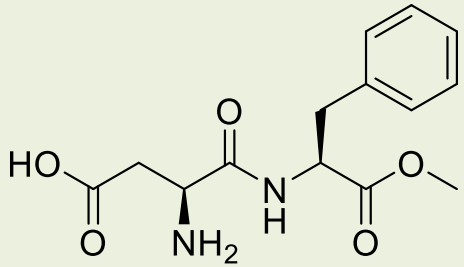
Saccharin

500 times sweeter



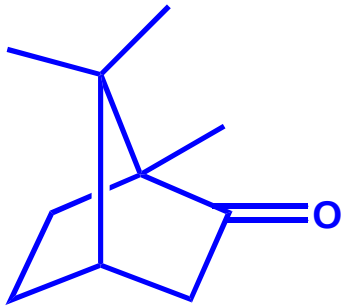
Cyclamate

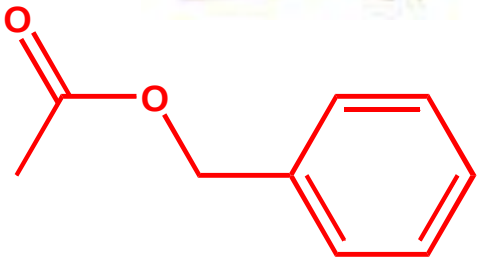
30 times sweeter



Aspartame

160 -200 times sweeter than natural sugar (sucrose)





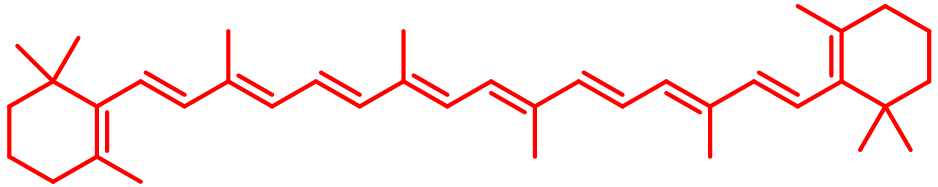
Camphor

1,7,7-Trimethylbicyclo[2,2,1]-heptan-2-one

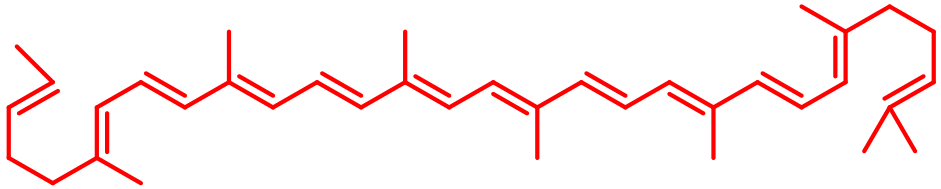
Incense

Benzyl acetate

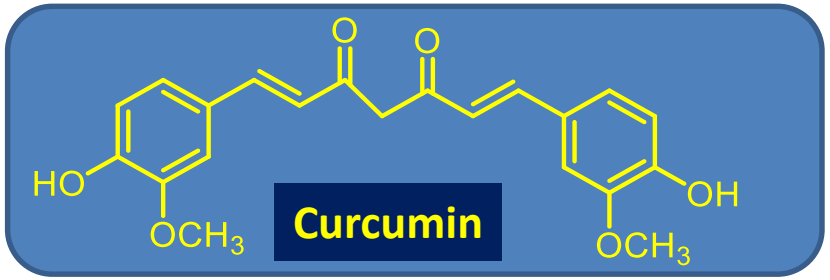
Fruits & Vegetables: Colour and Constitution

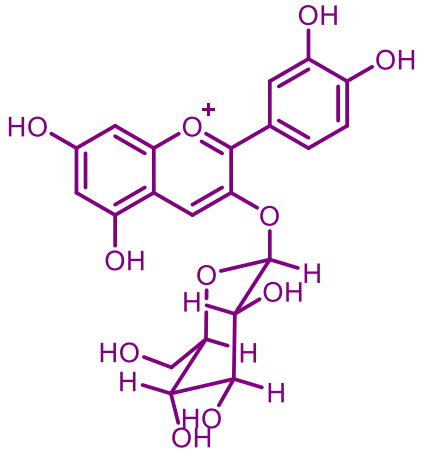


β-carotene (λ_{max} = 455 nm)



Lycopene, λ_{max} = 474 nm



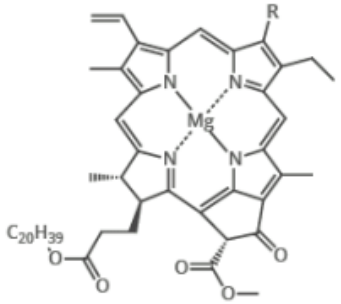


Anthocyanin derivative

THE CHEMISTRY OF BELL PEPPERS

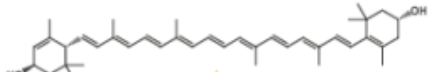
Bell peppers go through a spectrum of colours as they ripen – here we look at the compounds behind their colour, aroma, and flavour.

BELL PEPPER COLOUR CHEMISTRY

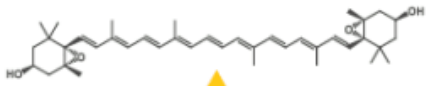


CHLOROPHYLL

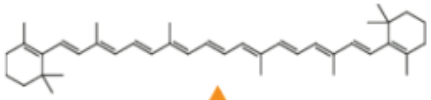
CHLOROPHYLL A: R = $-CH_3$
CHLOROPHYLL B: R = –CHO



LUTEIN

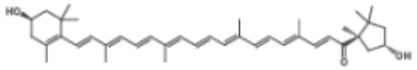


VIOLAXANTHIN

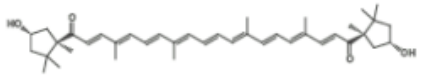


β-CAROTENE

Chlorophyll, used by plants for photosynthesis, gives bell peppers their initial green colour. As the pepper ripens, these are decomposed, and a range of carotenoid pigments appear. These include lutein, violaxanthin, and beta-carotene, which give yellow and orange hues. Eventually red carotenoid pigments including capsanthin and capsorubin appear. These red pigments are almost exclusively found in peppers.

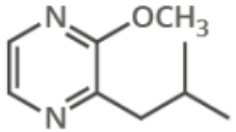


CAPSANTHIN

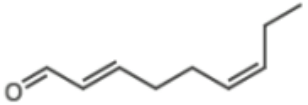


CAPSORUBIN

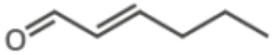
BELL PEPPER AROMA



BELL PEPPER PYRAZINE



CUCUMBER ALDEHYDE



(E)-2-HEXENAL

The aroma of bell peppers also develops as they ripen. In green peppers, the characteristic smell is largely due to a single chemical, 2-methoxy-3-isobutylpyrazine ("bell pepper pyrazine"). Other minor contributors include (E,Z)-2,6-nonadienal ("cucumber aldehyde"). The concentrations of most volatile compounds drop during ripening, with the exception of (E)-2-hexenal and (E)-2-hexenol, lending a sweeter, fruitier note to the aroma.

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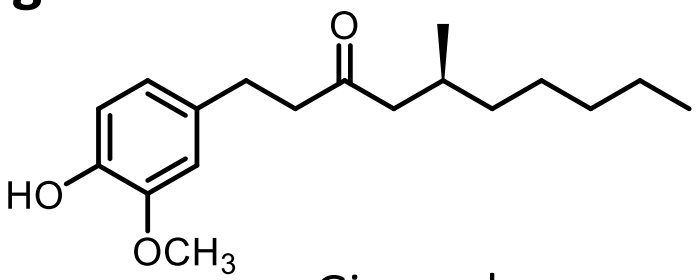
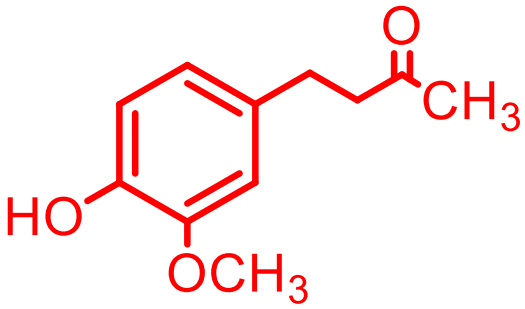
Chemistry of spices

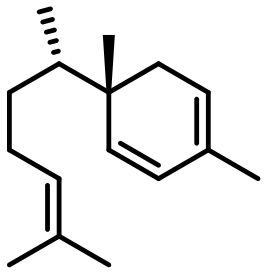
Ginger



Gingerol



Zingerone

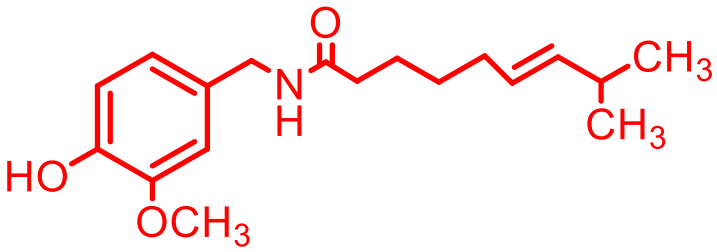


Zingiberene

Zingerone: The major pungent component in ginger. It has a role as an antioxidant, an anti-inflammatory agent, a radiation protective agent, an antiemetic, a flavouring agent and a fragrance.

Zingiberene gives ginger its distinct flavoring

Chille

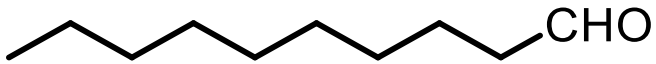


Capsaicin

Capsaicin is irritant to human and mammals. It cause burning sensation when tissues come in contact with it.

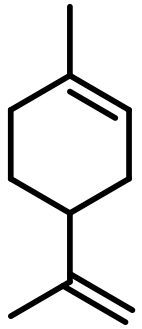
Chemistry of spices

Coriander

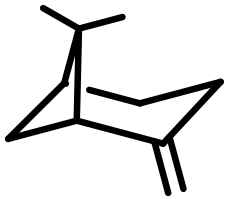


Decanal

Curry Leaves

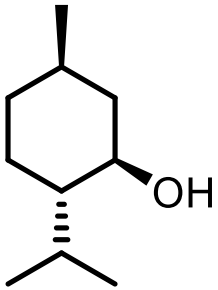


dipentene



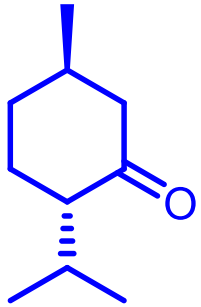
β-pinene

Mint



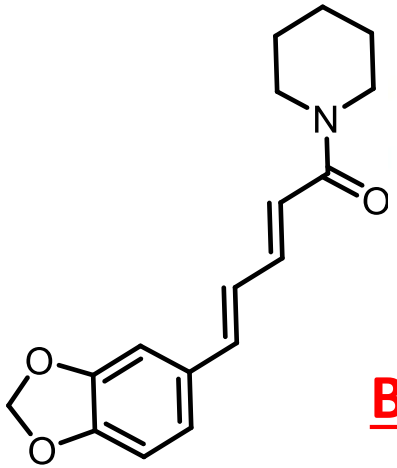
Menthol

Naturally it is available as (−)-menthol I1R,2S,5R)



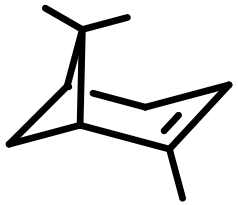
Menthone

Naturally it is available as (−)-menthone (2S,5R)



Pipreine

Black pepper

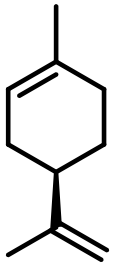


ά -pinene

Smell receptors

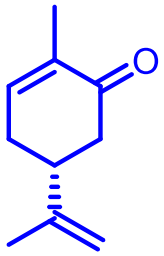
S-(-) Limonene

Pine or turpentine like odour



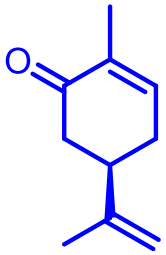
R-(+) Limonene

Citrus Flavour



R(-) carvone

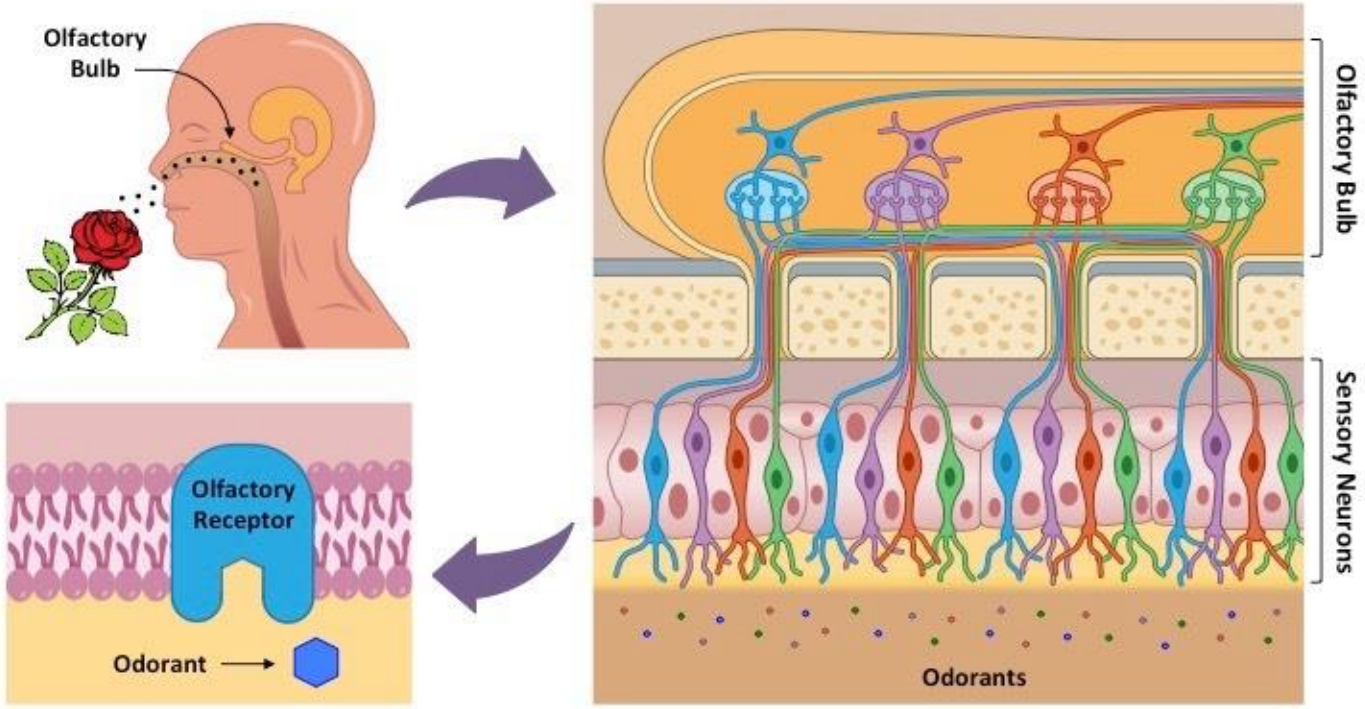
spearmint leaves smell



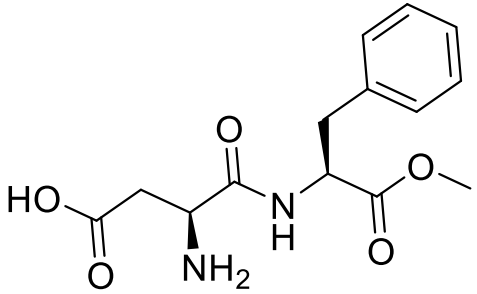
S(+) carvone

caraway (jeera) seed smell

Olfactory receptors (smell receptors) contain chiral groups which respond to each enantiomer differently.



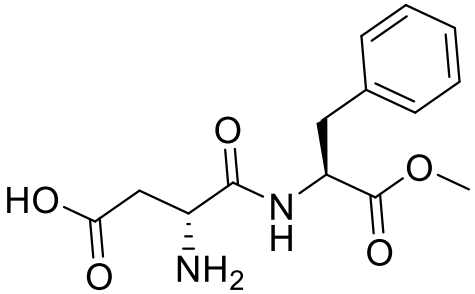
Tongue Receptors



(S,S) Aspartame

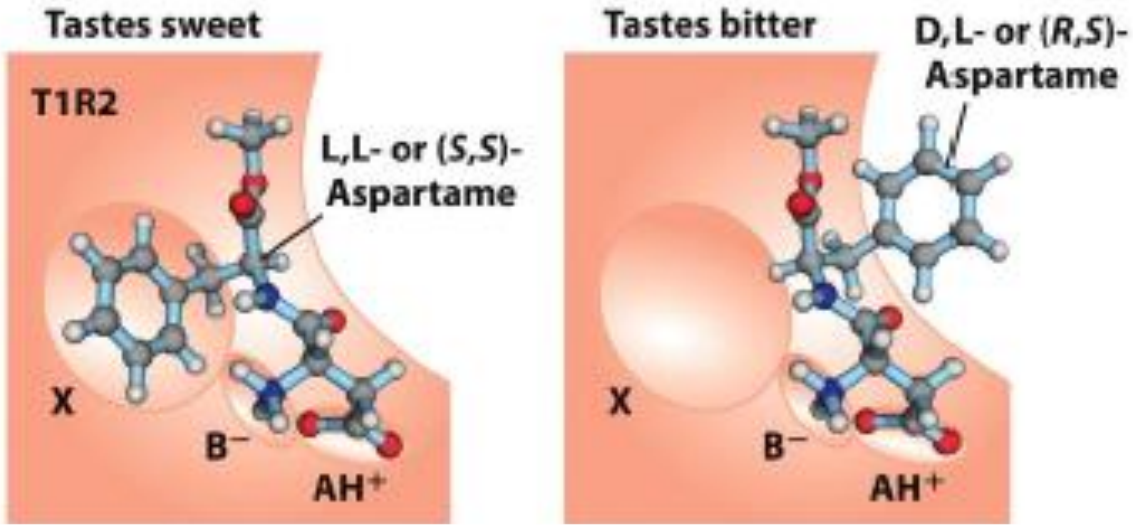
Methyl L-a-aspartyl-L-phenylalaninate

Sweet



(R,S) Aspartame

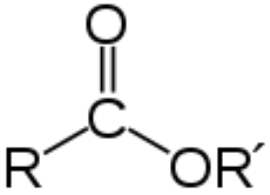
(Bitter)



Tongue receptors are also chiral and behave differently for two stereoisomers. For example (S, S) Aspartame and (R, S) Aspartame. The first one shows sweet taste and is approximately 200 times sweeter than sucrose.

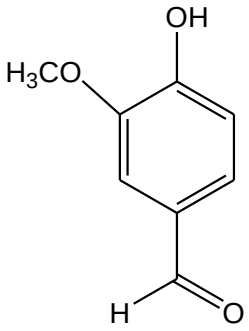
(S, S) Aspartame is used as artificial-synthetic sweetener, while the second one has bitter taste. One stereoisomer of Aspartame (S,S) (left) binds with high affinity to the receptors and triggers the sweet signal to the brain. The other stereoisomer (R, S) (right) does not bind to the sweetness receptor. This other stereoisomer actually binds to a bitterness receptor, and tastes bitter instead of sweet.

Fruity odours

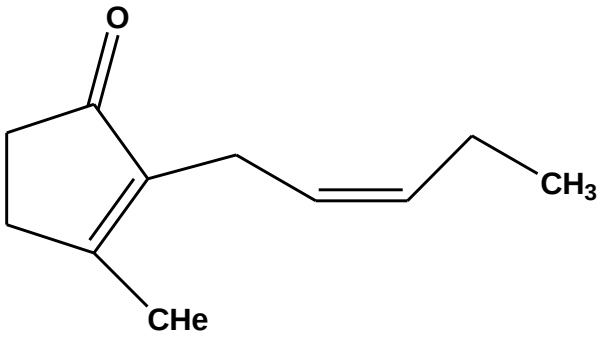


Chemical	Fruit Flavour
Butyl Butyrate; Allyl Hexanoate	Pineapple
Ethyl Butyrate	Banana, Pineapple, Strawberry
Ethyl nonanoate	Grape
Ethyl pentanoate	Apple

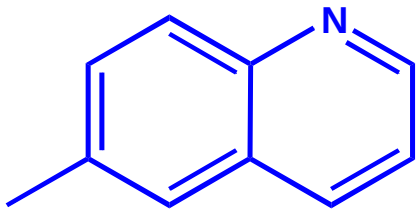
Flavours & Fragrances



Vanilla



Jasmone

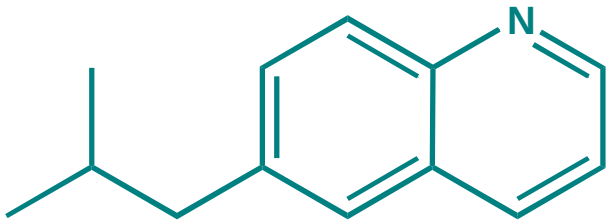


6-METHYLQUINOLINE

Blossom odour

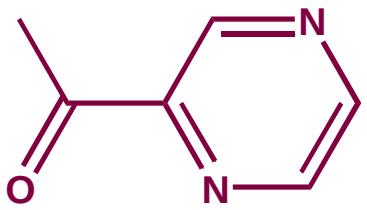
Lavender oil

ά -pinene, limonene,
1,8-cineole,
cis-ocimene,
trans-ocimene



6-ISOBUTYLQUINOLINE

Mossy earthy odour



2-ACETYLPYRAZINE

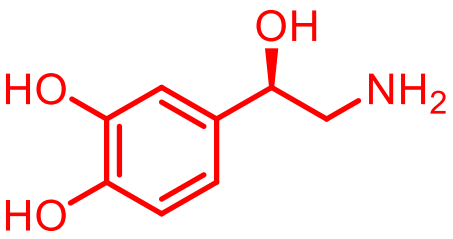
Popcorn like odour

Chemistry of Love

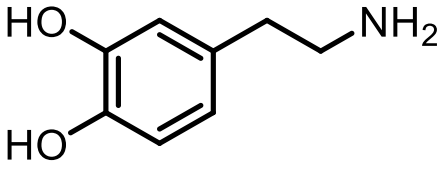
The emotions that you feel are a result of chemical messengers, primarily neurotransmitters.

Love, jealousy, envy, infatuation and infidelity all share a basis in chemistry.

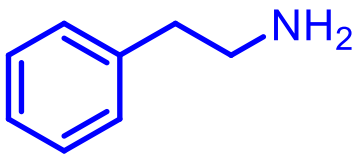
Chemistry of Emotions



Norepinepherine

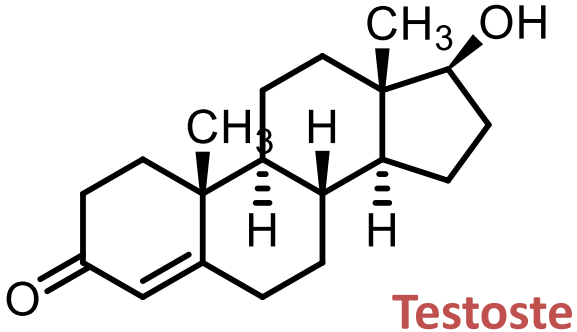


Dopamine



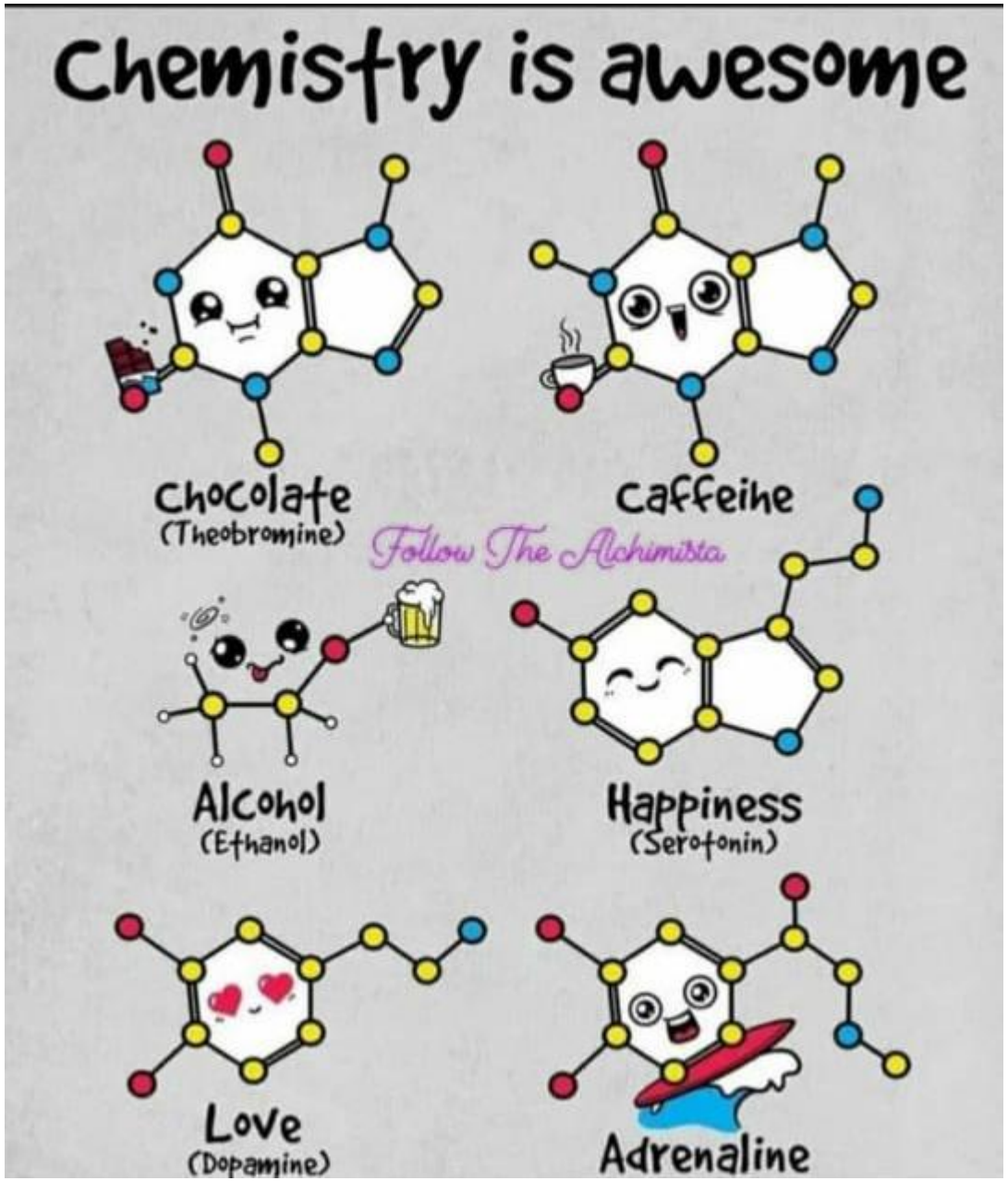
Phenylethylamine

- The sweaty palms and pounding heart of infatuation are caused by higher than normal levels of norepinephrine. It is also known as noradrenaline which is neurotransmitter and harmone which is Produced by adrenal glands. It is used to treat low blood pressure patients.
- Dopamine is known as feel good neurotransmitter.
- The 'high' of being in love is due to a rush of phenylethylamine and dopamine.



Testosterone

Did you know that raw lust is characterized by high levels of testosterone


Chemistry is awesome
Chocolate
(Theobromine)
Caffeine
Follow The Alchimista
Alcohol
(Ethanol)
Happiness
(Serotonin)
Love
(Dopamine)
Adrenaline